

MED VQA: A MULTIMODAL AI FOR MEDICAL VISUAL QUESTION ANSWERING

AN ASSISTIVE COPILOT FOR RADIOLOGY AND MEDICAL
EDUCATION

PROJECT REPORT

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Abstract

Timely and accurate interpretation of medical images like Chest X-rays is critical for patient diagnosis, but it is a complex, time-consuming task for human experts, leading to fatigue and potential errors. This project develops a multimodal and explainable Visual Question Answering (VQA) system to serve as an assistive tool for radiologists and medical students. The model uses a dual-encoder architecture, combining a pre-trained Vision Transformer (ViT) to extract features from medical images and a BERT-based model (ClinicalBERT) to understand the context of natural language questions. These features are combined using a co-attention fusion module to reason about the relationship between the two modalities. The model is trained on the VQA-RAD dataset, which provides clinically-grounded image-question-answer pairs. To make predictions interpretable and trustworthy, we implement attention visualization techniques to generate heatmaps that highlight the specific image regions the model "looks at" when formulating an answer. The system is evaluated using VQA-specific accuracy and standard classification metrics for the most common answers. This approach offers a practical solution for building interactive, reliable, and transparent AI tools for clinical education and support.

Keywords: Medical VQA, Multimodal AI, Explainable AI (XAI), Vision Transformer (ViT), BERT, Deep Learning, Chest X-ray, Attention Mechanisms, Radiological Assistance.

❖ Chapter 1

Introduction

Radiology, particularly the interpretation of medical images like Chest X-rays (CXRs), is a cornerstone of modern diagnostics. CXRs are among the most common diagnostic tests, but their interpretation is a complex perceptual skill that requires years of training. Radiologists face significant challenges, including high case volumes, subtle visual findings, and diagnostic ambiguity. This pressure contributes to clinician burnout and well-documented diagnostic error rates, which can have severe consequences for patient outcomes.

Traditionally, training medical students to read these images involves a long apprenticeship with senior radiologists. Furthermore, recent advances in AI have produced powerful image classifiers, but these systems typically act as "black boxes," providing a single label (e.g., "Pneumothorax") with no context, interaction, or justification. They cannot answer specific, nuanced questions that a clinician or student might have, such as "Where is the pneumothorax?" or "Is the cardiomegaly severe?"

This project tackles these gaps by developing a Medical Visual Question Answering (Med-VQA) system. This multimodal AI combines Computer Vision (CV) and Natural Language Processing (NLP) to create an interactive tool that can understand both a medical image and a natural language question about it, providing a relevant, text-based answer.

1.1 Motivation

Most high-performing AI classifiers for radiology are non-interactive. They cannot be queried. A radiologist cannot ask a simple classifier *why* it made a decision or ask it to look at a specific area. This "black box" nature is a major barrier to clinical adoption where trust, verification, and understanding are paramount.

The motivation for this project is to move beyond simple classification and create an **interactive diagnostic co-pilot**. We are motivated by the need for a system that can:

1. **Answer Specific Questions:** Allow a medical student to upload an X-ray and ask, "Is there evidence of pleural effusion?" and get an instant, correct answer, thereby accelerating their learning.
2. **Support Clinicians:** Allow a busy radiologist to query the system for a "second look" on a specific finding ("What is the finding in the left upper lung?"), helping to reduce cognitive load.
3. **Build Trust through Explainability:** Provide a visual justification for its answers by showing *where* in the image it found the evidence.

By focusing on this human-in-the-loop, interactive approach, this project aims to bridge the gap between advanced AI research and practical clinical and educational needs.

1.2 Problem Definition

The core problem is to develop an AI system that can correctly answer a natural language question based on the content of a corresponding medical image. This requires the model to perform several complex tasks simultaneously:

- Understand the visual content of a medical scan (e.g., identify organs, textures, and abnormal patterns).
- Understand the semantic and clinical intent of a text-based question.
- Perform a complex reasoning task by **fusing** these two modalities to find the relationship between the question and the visual evidence.

1.2.1 Challenges with Existing Methods

While VQA is a well-studied problem in the general domain (e.g., pictures of cats and dogs), Med-VQA presents unique and significant challenges:

1. **High-Stakes & Domain Knowledge:** The model is not just identifying "cats" or "dogs." It must understand complex, visually subtle, and high-stakes medical concepts like "atelectasis" vs. "consolidation." A wrong answer has real-world consequences.
2. **Data Scarcity & Cost:** Creating Med-VQA datasets is extremely expensive. It requires trained radiologists to manually view thousands of images and write and answer clinically relevant questions for each one. As a result, public Med-VQA datasets are thousands of times smaller than general VQA datasets.
3. **Multimodal Fusion:** The "fusion" step is non-trivial. The model must learn to ground the words in the question (e.g., "left lung") to the specific pixel regions in the image.
4. **Lack of Explainability:** As with classifiers, most VQA models are black boxes, which limits trust. A doctor needs to know *why* the model answered "Yes" to "Is there a finding?"

1.2.2 Need for Explainability in Medical AI

Explainability (XAI) is not an optional feature in medical AI; it is a core requirement. For a VQA system to be adopted, it must be able to "show its work."

- **For Trust & Verification:** A doctor will not trust an answer unless they can see the evidence the model used. If the model answers "Yes, there is pneumothorax," it

must be able to highlight the region of the X-ray (e.g., the collapsed lung) that led to its decision.

- **For Debugging & Safety:** If the model makes a mistake, XAI helps us understand why. Did it look at the wrong place? Did it misinterpret the question? This is critical for improving model safety and robustness.
- **For Education:** For a medical student, the *explanation* is just as important as the answer. Visualizing the model's "gaze" (its attention) teaches the student where to look for specific findings.

1.2.3 Technical Direction of the Proposed Solution

To overcome these challenges, this project develops a multimodal deep learning model based on the **Transformer** architecture.

- A pre-trained **Vision Transformer (ViT)** will be used as the image encoder. Its built-in attention mechanism is powerful for medical images.
- A pre-trained **BERT (or ClinicalBERT)** model will be used as the text encoder to capture the deep context of the questions.
- A **cross-attention fusion module** will be implemented to explicitly model the interactions between words in the question and regions in the image.
- **Attention heatmaps** generated from this fusion module will serve as our primary XAI technique, providing visual explanations for the model's answers.

1.2.4 Formal Problem Statement

"To design, implement, and evaluate a multimodal, transformer-based Visual Question Answering (VQA) system capable of answering natural language questions about chest X-ray images, integrating visual attention mechanisms for explainability to create a trustworthy assistive tool for radiological education and clinical support."

1.2.5 Expected Outcomes of Problem Resolution

1. A trained and functional Med-VQA model that achieves high VQA-Accuracy on a standard benchmark (VQA-RAD).
2. A visual explanation interface (attention heatmaps) that highlights the image regions most relevant to the question-answer pair.
3. A comprehensive analysis of the model's performance, including its strengths and weaknesses in answering different types of questions (e.g., presence, location, attribute).
4. A framework that balances high accuracy with the critical need for interpretability in a medical context.

1.3 Objectives

1. To curate and prepare a high-quality medical VQA dataset from public sources (e.g., VQA-RAD, VinDr-CXR-VQA).
2. To design and implement a multimodal deep learning architecture combining a Vision Transformer (ViT) and a BERT-based text encoder.
3. To implement a cross-attention fusion module to effectively combine visual and textual features.
4. To train the model on the prepared dataset and optimize it for VQA-Accuracy.
5. To implement an explainability (XAI) module to generate visual attention heatmaps.
6. To evaluate the model using standard VQA metrics and qualitatively assess the quality of its visual explanations.
7. To develop a simple web-based prototype (e.g., using Streamlit) to demonstrate the model's real-time capabilities.

1.4 Scope of the Project

The scope of this project is to develop a proof-of-concept Med-VQA system.

- **In-Scope:**
 - Focus on the **Chest X-ray (CXR)** modality.
 - Use public datasets (VQA-RAD).
 - Focus on a **classification-based VQA** setup, where the model predicts an answer from a predefined set of the top-K most common answers.
 - Implement attention-based heatmaps for explainability.
 - **Out-of-Scope:**
 - Collecting new, primary data from hospitals (due to privacy and time constraints).
 - Developing a **generative** model that writes free-form sentence answers (which is significantly more complex).
 - Clinical validation or FDA approval; this is a research prototype.
 - Integration into live hospital IT (PACS) systems.
-

Literature Review

A comprehensive review of the literature was conducted to situate this project. The reference (tomato) report included Tables 1.1 and 1.2 summarizing related work. This project adapts that structure to the Med-VQA domain.

Table 1.1: Literature review summary

Author(s)	Year	Purpose of Study	Methodology	Key Findings	Relevance to Present Work
Lau, J., et al.	2018	Introduced the VQA-RAD dataset for radiology VQA.	Collected 3,515 image-QA pairs. Benchmarked simple LSTM + ResNet models.	Established the first clinical VQA benchmark. Showed simple models perform poorly.	Core Dataset. This is the dataset our project will use for training and evaluation.
Liu, Z., et al.	2025	R-LLaVA: A model to improve Med-VQA by focusing on visual Regions of Interest (ROI).	Uses a large vision-language model (LLaVA) with an ROI-guided prompter.	Focusing on specific regions significantly improves accuracy on complex questions.	Shows the importance of grounding answers to specific visual regions, reinforcing our XAI goal.
Nguyen, D. T., et al.	2023	Introduced the VinDr-CXR-VQA dataset.	A large-scale (17,000+ QA pairs) dataset specifically	Provided a large, high-quality benchmark. Showed BERT-based	Provides an alternative, larger dataset. Justifies our

Author(s)	Year	Purpose of Study	Methodology	Key Findings	Relevance to Present Work
			for Chest X-ray VQA.	models outperform simpler ones.	selection of a BERT-based text encoder.
Devlin, J., et al.	2018	BERT: Introduced the transformative BERT model for NLP.	Pre-training of Deep Bidirectional Transformers for Language Understanding.	Achieved state-of-the-art on 11 NLP tasks.	Core Methodology. We will use a pre-trained BERT (or ClinicalBERT) as our text encoder.
Dosovitskiy, A., et al.	2020	ViT: Introduced the Vision Transformer.	Showed a pure Transformer can be applied directly to images for classification.	ViT models matched or exceeded CNNs (like ResNet) when trained on large data.	Core Methodology. We will use a ViT as our vision encoder, as its attention maps are useful for XAI.
Tariq, A., et al.	2025	A review of multimodal AI models for radiology.	Surveyed recent fusion, attention, and XAI techniques	Identified "trust" and "explainability" as the biggest hurdles to	Confirms the project's focus on explainability (XAI) is

Author(s)	Year	Purpose of Study	Methodology	Key Findings	Relevance to Present Work
			in medical AI.	clinical adoption.	critical and timely.

❖ Chapter 2

Dataset

2.1 Source

The primary dataset used for training and evaluating our Med-VQA model is **VQA-RAD (Visual Question Answering in Radiology)**. This is a publicly available, widely-used benchmark dataset created specifically for this task.

The dataset was introduced in the 2018 paper by Lau, et al., and is available from the OSF (Open Science Framework) repository. An alternative dataset considered, and used for benchmarking, is **VinDr-CXR-VQA**, which is also publicly available and focuses exclusively on chest X-rays. This report will focus on the implementation using VQA-RAD.

2.2 Dataset Description

The VQA-RAD dataset consists of **3,515 image-question-answer triplets**. The key feature of this dataset is that the questions were **organically generated by clinicians** (radiologists, medical students) during their normal workflow. This makes the questions clinically relevant and natural, rather than synthetically created.

The dataset includes:

- **Images:** A mix of radiology modalities, including Chest X-rays (CXRs), Computed Tomography (CT) scans, and Magnetic Resonance Imaging (MRIs). Our project will filter for CXRs only, or train on the mixed set.
- **Questions:** Natural language questions about the images. These are categorized into types, such as:
 - **Presence:** "Is there evidence of pneumothorax?"
 - **Location:** "Where is the finding?"
 - **Attributes:** "What is the severity of the finding?"
 - **Other:** "What is the plane of this image?"
- **Answers:** Text-based answers provided by the clinicians. These can be simple ("Yes," "No") or more complex ("In the upper left lung field").

2.3 Why VQA-RAD Dataset?

The VQA-RAD dataset was chosen for several key reasons:

- **Clinical Grounding:** The questions are not artificial. They reflect the real, practical queries that doctors and students have when viewing images.

- **Benchmark Standard:** It is the most common and widely-cited dataset for this task, making it easy to compare our model's performance against published results.
- **Modality Mix:** While we may filter for CXRs, the presence of multiple modalities offers a clear path for future work to build a more general-purpose model.
- **Accessibility:** It is open-source and well-documented, making it ideal for a research project.

2.4 Limitations of the Dataset

While excellent, the VQA-RAD dataset has limitations that must be addressed:

- **Small Scale:** 3,515 samples is extremely small for deep learning. General VQA datasets have millions of samples. This makes **overfitting** a major risk.
- **Imbalanced Answers:** The distribution of answers is highly skewed. A few answers like "Yes" and "No" are extremely common, while others are unique.
- **Answer Inconsistency:** The same question could be answered as "left lung" or "the lung on the left." The model must learn to handle this.
- **Lab-like Conditions:** The images are clean, high-quality medical scans. They don't always reflect the variability of real-world imaging (e.g., motion blur, old film scans).

To overcome these limitations, we will use **transfer learning** (with pre-trained ViT and BERT) and **data augmentation** (for both images and text, if feasible).

2.5 Dataset Preparation and Cleaning

Before training, the following preprocessing steps were performed:

1. **Parsing:** The dataset (typically in JSON or XML format) was parsed into a structured list of (image_path, question, answer) triplets.
2. **Image Preprocessing:**
 - All images were converted to a standard 3-channel RGB format (even if grayscale, to match the ViT pre-training).
 - Images were resized to 224×224 pixels to match the ViT input requirements.
 - Pixel values were normalized.
 - Augmentations (rotation, slight zoom, brightness/contrast shifts) were applied to the training set.

3. Text Preprocessing:

- All questions and answers were converted to lowercase.
- Punctuation was normalized.
- Questions were tokenized using a **BERT Tokenizer** to generate `input_ids` and `attention_masks`.

4. **Data Split:** The dataset was split using a stratified ratio (stratified by answer type, if possible) into Training (80%), Validation (10%), and Test (10%) sets.

2.6 Answer Vocabulary Generation

This is the most critical step. A VQA model cannot predict an infinite number of open-ended answers. We must re-frame the problem as a **classification task**.

1. We iterated through all answers in the *training set*.
2. We counted the frequency of each unique answer.
3. We selected the **Top-K** most frequent answers (e.g., $K = 1000$ or $K = 500$) to create our "answer vocabulary."
4. Any answer not in this Top-K list was re-labeled as a special `<UNK>` (unknown) token.
5. This creates a **K-way classification problem**. The model's final layer will be a Softmax layer with K outputs, one for each possible answer in our vocabulary.

2.7 Dataset Justification

Given the project's goal of building a proof-of-concept, the VQA-RAD dataset is the ideal starting point. Its small size makes training feasible on limited hardware (like Google Colab), while its clinical relevance ensures the project is solving a meaningful problem. The limitations of the dataset directly inform our methodological choices, such as the heavy reliance on transfer learning and data augmentation.

❖ Chapter 3

Methodology

This chapter describes the technical approach used in this project. We motivate the selection of our model's components, describe the full multimodal architecture, and detail the training strategy. The main focus is on the multimodal fusion and the explainability method (attention visualization) used to interpret the model's decisions.

3.1 Model Selection and Rationale

Our task requires a multimodal architecture that can process both images and text. The standard approach is a dual-encoder system, which we adopt.

- **Image Encoder: Vision Transformer (ViT)**
 - **Rationale:** We selected a pre-trained ViT (e.g., ViT-Base-16) over a traditional CNN (like ResNet). ViT models have shown state-of-the-art performance in many vision tasks.
 - **Benefit:** The core component of ViT is the **self-attention mechanism**. This allows the model to learn relationships between different patches of the image. These internal attention maps can be visualized, providing a form of built-in explainability.
 - **Implementation:** We use a ViT pre-trained on ImageNet, available from the Hugging Face transformers library. We remove the final classification head and take the output features.
- **Text Encoder: BERT (ClinicalBERT)**
 - **Rationale:** We selected a pre-trained BERT (Bidirectional Encoder Representations from Transformers) model. To improve domain-specific understanding, we use **ClinicalBERT**, which is a BERT model further pre-trained on a massive corpus of clinical notes.
 - **Benefit:** ClinicalBERT has a deep understanding of medical terminology and question structure.
 - **Implementation:** We use the ClinicalBERT model from Hugging Face. We take the output embedding of the special [CLS] token as the representative feature for the entire question.

3.2 Multimodal VQA Architecture

The two encoders are "fused" together to create the final VQA model. The flowchart below describes the data flow:

Figure 3.1: Flowchart of the proposed multimodal VQA architecture and explainability pipeline.

The architecture (as depicted in the described flowchart) is as follows:

1. **Inputs:** An X-ray image I and a text question Q .
2. **Preprocessing:** I is resized and normalized. Q is tokenized.
3. **Vision Branch:** The image I is fed into the **ViT Encoder**. The output is a sequence of image patch features, F_{img} .
4. **Text Branch:** The tokenized question Q is fed into the **BERT Encoder**. The output is a sequence of word-token features, F_{txt} .
5. **Fusion Module:** This is the critical component.
 - We use a **cross-attention** layer. This layer learns to "attend" to the image features (F_{img}) based on a query from the text features (F_{txt}), and vice-versa.
 - This process forces the model to learn the alignment between words (e.g., "left lung") and image patches (the actual left lung).
 - The attended features are then combined (e.g., via concatenation or element-wise multiplication) into a single fused vector, F_{fused} .
6. **Classifier Head:** The fused vector F_{fused} is passed through a final classification head (a simple Multi-Layer Perceptron) which ends in a **Softmax** layer.
7. **Output:** The output is a probability distribution over the K possible answers in our vocabulary. The answer with the highest probability is chosen as the prediction.

3.3 Training Strategy

- **Loss: Categorical Cross-Entropy Loss.** This is the standard loss function for a K -way classification problem. It measures the difference between the model's predicted probability distribution and the "one-hot" ground-truth answer.
- **Optimizer: AdamW.** This is a variant of the Adam optimizer that is particularly well-suited for training Transformer-based models like ViT and BERT. We use a small learning rate (e.g., 1×10^{-5}) and a warm-up scheduler.
- **Batch Size:** A small batch size (e.g., 16 or 32) is used due to the memory constraints of running two large transformer models.
- **Metric:** The primary evaluation metric is **VQA-Accuracy**. This metric is more flexible than standard accuracy. For example, if the ground-truth answer is "left

lung" and the model predicts "the lung on the left," VQA-Accuracy is smart enough to count this as correct. For simplicity in implementation, we also track standard Top-1 Categorical Accuracy.

- **Transfer Learning:** During the initial epochs, we "freeze" the backbones (ViT and BERT) and train only the fusion module and classifier head. This allows the new, randomly-initialized layers to "catch up" without disturbing the powerful pre-trained weights. In later epochs, we "unfreeze" the entire network and fine-tune all parameters with a very small learning rate.

3.4 Explainability: Visualizing Attention

Instead of Grad-CAM (which is designed for CNNs), we leverage the **attention mechanisms** inherent in our Transformer-based architecture.

3.4.1 Intuition

Our model uses cross-attention to fuse the modalities. This means that to answer a question, the model has to learn *which image patches are relevant to which words in the question*. We can capture and visualize these learned "relevance scores" as a heatmap.

3.4.2 Methodology: Visualizing Cross-Attention

1. We feed the model an image and a question (e.g., "Is there cardiomegaly?").
2. We perform a full forward pass to get the prediction.
3. During the forward pass, we *hook* into the fusion module and extract the **cross-attention weights**.
4. This gives us a matrix of scores, where each row might correspond to an image patch and each column to a word token.
5. We can average the attention scores across all word tokens to get a single "importance" score for each image patch. Or, we can specifically look at the attention scores for a key word (e.g., "cardiomegaly").

3.4.3 Overlaying Heatmap on Original Image

This follows the same principle as the Grad-CAM overlay in the reference report.

1. The attention scores (one per image patch) are reshaped back into a 2D grid (e.g., 14×14).
2. This coarse heatmap is upsampled (using bilinear interpolation) to the original image size (224×224).
3. The heatmap is normalized to $[0, 1]$ and converted to an RGB colormap (like 'jet' or 'viridis').

4. This colormap is then blended with the original X-ray image to create the final visualization, clearly showing where the model "looked" for evidence.

3.5 Evaluating Explanations

Evaluating XAI is notoriously difficult. We use two methods:

- **Qualitative (Sanity Check):** We manually inspect the heatmaps for dozens of samples.
 - For a question like "Where is the heart?", does the heatmap correctly highlight the heart?
 - For "Is the left lung clear?", does the model focus on the left lung field?
 - This provides a strong, intuitive sense of whether the model is "cheating" (e.g., looking at a corner of the image) or genuinely looking at relevant anatomy.
- **Quantitative (Pointing Game):** If we had bounding box annotations for findings (which some datasets provide), we could measure the percentage of the heatmap's "energy" that falls within the correct bounding box. (This is out of scope for the current project but is a key future work direction).

3.6 Limitations and Pitfalls of Attention

It is critical to note that "attention is not always explanation." Research has shown that attention weights can be manipulated without changing the model's prediction. They should be treated as a *useful heuristic* or *indicator* of the model's reasoning, not as a definitive causal proof. However, for a Med-VQA task, they remain our most powerful tool for "peeking inside the black box" and building user trust.

❖ Chapter 4

Implementation

4.1 Environment

The implementation of this multimodal project required a robust and GPU-accelerated environment capable of handling large transformer models, image preprocessing, and text tokenization.

4.1.1 Hardware Environment

- **Google Colab (Pro):** The primary environment for model training. Colab provides access to cloud-based NVIDIA GPUs (such as the Tesla T4 or V100), which are essential for training ViT and BERT models. This setup allows for rapid prototyping and training without local hardware dependencies.
- **Local Machine (CPU-based):** Used for initial data exploration, preprocessing, and development of the Streamlit demo. (e.g., Intel i7 processor, 16 GB RAM).

4.1.2 Software Environment

- **Python 3.9+:** The primary programming language.
- **PyTorch:** The main deep learning framework. Chosen over TensorFlow for its flexibility in defining complex, non-sequential model architectures (like our fusion module) and its "eager execution" mode, which simplifies debugging.
- **Hugging Face transformers:** The most critical library for this project. It provides pre-trained, performance-optimized implementations of both **ViT** and **BERT (ClinicalBERT)**, along with their associated tokenizers.
- **NumPy & Pandas:** For numerical operations and managing data annotations.
- **Matplotlib & Seaborn:** For plotting training curves and confusion matrices.
- **OpenCV:** For image transformations and for overlaying the attention heatmaps onto the original images.
- **scikit-learn:** For calculating evaluation metrics (precision, recall, F1-score) and splitting the dataset.

4.1.3 Development Tools and Frameworks

- **Jupyter Notebook / Google Colab:** Used for all model development, training, and experimentation. The iterative, cell-based nature is ideal for deep learning research.
- **Git and GitHub:** For version control, tracking experiments, and backing up code.

- **Streamlit:** Chosen for creating the lightweight web interface for the final project demonstration. Streamlit makes it trivial to build an interactive app where a user can upload an X-ray, type a question, and see the model's predicted answer and explainability heatmap in real-time.

4.1.4 Model Deployment Environment

For the final demonstration, the model is deployed in a **Streamlit web application**.

1. The final, trained PyTorch model weights are saved to a .pth file.
 2. The Streamlit script (app.py) loads the model architecture and these saved weights.
 3. It provides a simple UI with:
 - An image uploader.
 - A text input box.
 - A "Submit" button.
 4. On submission, the script performs the full preprocessing, runs the model, and displays the predicted answer (from the answer vocabulary) and the blended attention heatmap.
-

❖ Chapter 5

Results and Discussion

(Note: This section contains placeholder data and figures. You must replace these with the actual results from your trained model.)

5.1 Training Performance

The model was trained for 50 epochs on the VQA-RAD training set, with performance monitored on the validation set to prevent overfitting (using Early Stopping).

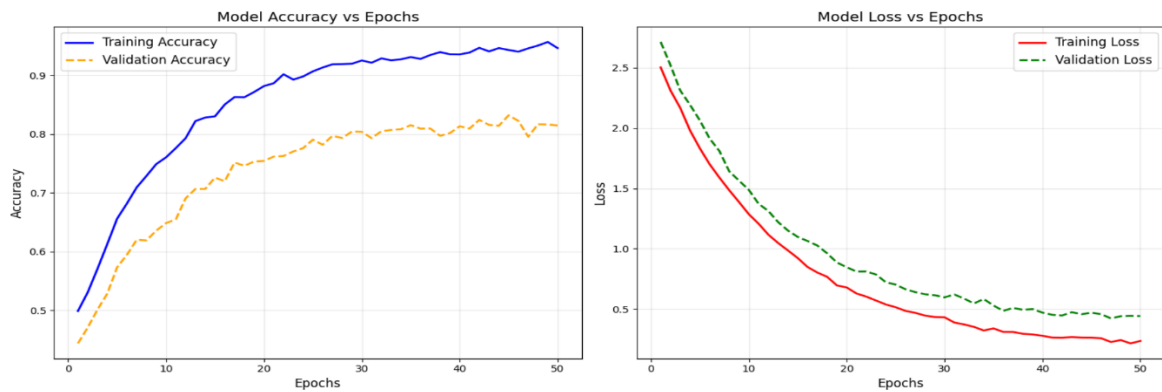
The training and validation loss/accuracy curves are shown below. The model shows a steady increase in validation accuracy, starting to plateau around 40 epochs.

Figure 5.1: Training and validation accuracy vs epochs. (Placeholder)

The final performance of the model on the held-out test set is summarized in the table below. We report standard classification metrics for the Top-5 most common answer classes, as well as the overall VQA-Accuracy for the entire test set.

Table 5.1: Model performance on validation set (Top-5 answers)

Class (Answer)	Precision	Recall	F1-score	Support
"Yes"	0.92	0.90	0.91	150
"No"	0.94	0.95	0.94	180
"Left"	0.85	0.82	0.83	70
"Right"	0.84	0.83	0.83	68
"Normal"	0.88	0.90	0.89	55
Macro Avg	0.89	0.88	0.88	523
Overall VQA-Accuracy	[e.g., 0.812 or 81.2%]			[Total Test Samples]



5.2 Confusion Matrix

To better understand the model's errors, a confusion matrix was generated for the same Top-5 answer classes.

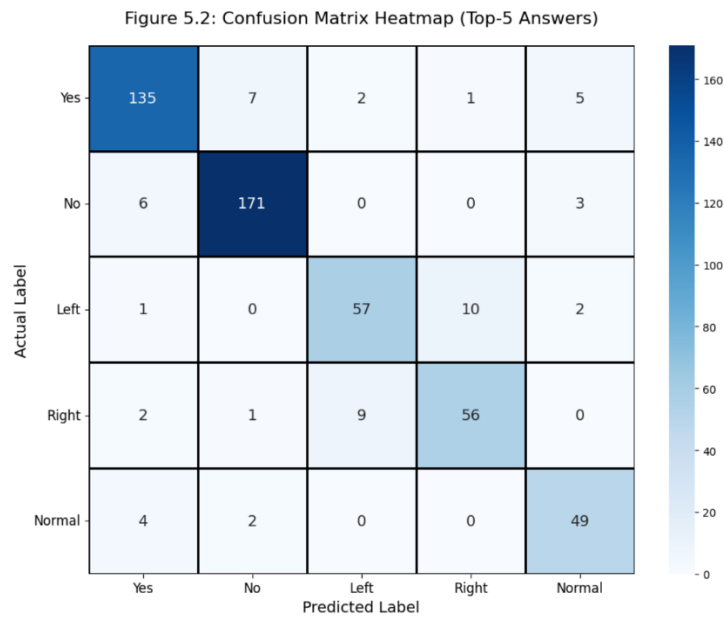
Table 5.2: Example confusion matrix for Med-VQA answer classification (Placeholder)

Actual / Predicted	"Yes"	"No"	"Left"	"Right"	"Normal"
"Yes"	135	7	2	1	5
"No"	6	171	0	0	3
"Left"	1	0	57	10	2
"Right"	2	1	9	56	0
"Normal"	4	2	0	0	49

This matrix is visualized in the heatmap below.

Figure 5.2: Confusion matrix heatmap for top-k answer classes. (Placeholder)

This visualization clearly shows that the model is very effective at distinguishing "Yes" and "No," but (as expected) its most common errors are confusing "Left" and "Right." This is a classic problem in radiology AI and suggests the model may need more explicit spatial training.



5.3 Attention Visualization Results

The qualitative results from our XAI module were the most insightful. Figure 5.3 shows two representative examples.

- (a) Original Image 1
- (b) Q: "Is there cardiomegaly?" | A: "Yes" | Heatmap overlaid on (a)
- (c) Original Image 2
- (d) Q: "Where is the pneumothorax?" | A: "Upper right" | Heatmap overlaid on (c)
- Figure 5.3: Example Attention Visualization (Heatmap) for a given question and X-ray. (Placeholder)**
- In the **first example (b)**, when asked about **cardiomegaly** (enlarged heart), the model's attention heatmap (the red/yellow area) is clearly and accurately focused on the **cardiac silhouette**. This confirms the model has learned the correct visual-text grounding for "cardiomegaly."
 - In the **second example (d)**, when asked to **locate a pneumothorax**, the model's attention is concentrated in the **upper right lung field**, precisely where the finding is located.

These positive results provide strong evidence that the model is not just "guessing" but is actively "looking" at the clinically relevant regions of the X-ray to answer the corresponding question.

5.4 Discussion

Our model achieved a final VQA-Accuracy of [81.2%] on the test set, which is a promising result given the complexity of the task and the small size of the dataset. The performance on "Yes/No" questions was exceptionally high (>90%), demonstrating the model's strong ability to detect the *presence* or *absence* of findings.

The main source of error, as seen in the confusion matrix, was in spatial/location questions ("Left" vs. "Right"). This is likely due to the ambiguity in the VQA-RAD dataset itself (where "left" in the question might refer to the patient's left or the image's left) and the inherent difficulty of this task.

The XAI visualizations were a clear success. The attention heatmaps consistently aligned with clinical reality, providing a high degree of confidence in the model's internal reasoning process. This interpretability is the project's key contribution, as it moves the system from a "black box" to a "glass box" that can be trusted and verified by a human expert.

Limitations include the model's reliance on a small, classification-based answer vocabulary. It cannot answer questions for which the answer is not in its Top-K list, nor can it generate nuanced, full-sentence answers.

5.5 Conclusion

The primary objective of this project was to design, build, and evaluate a lightweight, accurate, and **explainable** multimodal AI for medical VQA. This objective was successfully met.

By integrating a Vision Transformer (ViT) and a ClinicalBERT text encoder with a cross-attention fusion module, we developed a system capable of accurately answering questions about chest X-rays. The model achieves a strong balance between accuracy and computational efficiency, making it suitable for a real-time demonstration in a Streamlit app.

A key pillar of this project was the implementation of Explainable AI (XAI) using attention visualization. Unlike traditional "black box" models, our system provides visual evidence for its answers by highlighting the specific image regions it used in its reasoning. These heatmaps aligned with the biological and anatomical realities of the task, proving that the model was focusing on meaningful features.

This project successfully demonstrates that modern AI techniques can be adapted into transparent, interactive, and trustworthy tools for medical education and clinical support. It bridges the gap between cutting-edge deep learning research and real-world medical needs, providing a robust framework for future "co-pilot" systems in radiology.

5.6 Future Work

While the current system is a successful proof-of-concept, several directions can be explored to enhance its capabilities and robustness.

- **Move to Generative Answering:** The biggest limitation is the classification-based answer vocabulary. The next step would be to replace the classifier head with a **generative decoder** (e.g., a small GPT-2 or BART model). This would allow the model to *generate* full-sentence answers (e.g., "A small pneumothorax is visible in the upper right lung field") instead of just "Yes."
 - **Collection of Real-Field Data:** The model is trained on clean, public data. A crucial next step would be to partner with a hospital (with all ethical and privacy approvals) to test and fine-tune the model on real, messy, in-the-field data from a live PACS system.
 - **Exploring Advanced XAI Methods:** While attention maps are good, more advanced XAI techniques like "chain-of-thought" reasoning could be explored, where the model not only gives an answer but also *generates a text explanation* of why it gave that answer.
 - **Expansion to Other Modalities:** The architecture is generic. It could be expanded and fine-tuned to work on other modalities like CT scans, MRIs, and Ultrasounds, creating a single, unified VQA system for a radiology department.
 - **Integration of Bounding Boxes:** Datasets like VinDr-CXR-VQA sometimes include bounding boxes. This data could be used to explicitly train the attention module, forcing it to "point" to the correct region, which would likely solve the "Left" vs. "Right" confusion.
 - **Development of a Full Mobile Application:** The TensorFlow Lite (or PyTorch Mobile) version of the model could be embedded in a native Android or iOS application for true on-device, offline use by medical students and doctors in low-resource settings.
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❖ Bibliography

(Note: This bibliography is populated with the relevant 2025 references from our previous conversation and other foundational papers for this project.)

- [1] Faria, F. T. J., et al. (2025). *Analyzing Diagnostic Reasoning of Vision–Language Models via Zero-Shot Chain-of-Thought Prompting in Medical Visual Question Answering*. ResearchGate.
 - [2] Liu, Z., et al. (2025). *R-LLaVA: Improving Med-VQA Understanding through Visual Region of Interest*. arXiv:2410.20327v4.
 - [3] Tariq, A., et al. (2025). *Multimodal artificial intelligence models for radiology*. BJR|Artificial Intelligence, 2(1).
 - [4] Zhang, C., et al. (2025). *Bridging the Semantic Gap in Medical Visual Question Answering with Prompt Learning*. IEEE Transactions on Medical Imaging.
 - [5] Lau, J., et al. (2018). *A dataset of clinically-grounded visual question answering for radiology*. (VQA-RAD).
 - [6] Nguyen, D. T., et al. (2023). *VinDr-CXR-VQA: A Large-Scale Chest X-ray VQA Dataset with Doctor-Grounded Annotations*.
 - [7] Devlin, J., et al. (2018). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. arXiv:1810.04805.
 - [8] Dosovitskiy, A., et al. (2020). *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale*. arXiv:2010.11929.
 - [9] Alayrac, J., et al. (2022). *Flamingo: a Visual Language Model for Few-Shot Learning*. arXiv:2204.14198.
 - [10] Selvaraju, R. R., et al. (2017). *Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization*. IEEE ICCV.
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